

ELHARRAR Mehdi

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Education

Master in Public Health (MPH) Institut Supérieur des Sciences de la Santé – Université Hassan Premier	2024 – 2026 Settat, Morocco
Bachelor's Degree in Nursing Institut Supérieur des Sciences Biologiques et Paramédicales – UM6P	2021 – 2024 Benguerir, Morocco

Research Experience

Master's Research Intern – Epidemiology Faculty of Medical Sciences (FMS), UM6P	Mar 2026 – Jun 2026 Benguerir, Morocco
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Dissertation: Prevalence and Determinants of Non-Communicable Disease–Related Multimorbidity in Africa: A Harmonized Analysis of the WHO STEPS Surveys

Supervisors: Prof. Nasser LAOUALI & Dr. Asmaa HABIB

- Estimated the prevalence and composition of cardiometabolic multimorbidity (hypertension, diabetes, hypercholesterolemia, cardiovascular disease, obesity) across 40 African countries using nationally representative WHO STEPS surveys.
- Harmonized and standardized 55 large-scale STEPS datasets (219,197 participants), building a reproducible workflow to reconcile heterogeneous variable definitions, units, and measurement protocols across surveys.
- Mapped cross-country geographic variation and characterized the most frequent disease co-occurrences using age-standardized prevalence estimates.
- Built multivariable regression models to identify sociodemographic and behavioral determinants, and to quantify the extent to which risk factors are shared across conditions.

Undergraduate Research Intern – Nursing CHU Mohamed VI	Apr 2024 – Jun 2024 Marrakesh, Morocco
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Dissertation: Moral sensitivity: what role in ethical decision-making in nursing?

Supervisor: Prof. Majda SEBBANI

- Conducted a quantitative study evaluating the influence of moral sensitivity on ethical clinical decision-making within hospital settings.
- Designed the study protocol, executed primary data collection, and performed descriptive statistics and hypothesis testing using SPSS.

Independent Quantitative Projects

Self-directed projects, delivered as reproducible R Markdown reports; computational notebooks and full analytical outputs available on my [open digital portfolio](#).

- **Non-parametric inference & reproducible scientific writing:** Applied bootstrap resampling in R to construct confidence intervals for median differences where parametric assumptions were unmet; drafted a reproducible manuscript in the Elsevier LaTeX template with BibTeX reference management.
- **Longitudinal modeling :** Modeled repeated-measures biomarker trajectories (blood pressure, cholesterol) across visits using linear mixed-effects models to account for within-subject correlation; complemented with survival analysis to link baseline risk factors to cardiovascular events.

- **Survival analysis of child mortality:** Modeled demographic and socioeconomic determinants of infant and juvenile survival from fertility-survey data using Kaplan–Meier estimation and multivariable Cox proportional-hazards regression.

Skills and Tools

- **Methodological skills:** Epidemiological study design; multivariable regression modeling; survival analysis; survey analysis; data management and harmonization.
- **Statistical Software & Programming :** R, SPSS, SQL.
- **Technical Tools:** Microsoft Office Suite, LaTeX, Git/GitHub.
- **Languages:** English & French (Professional Working Proficiency).

Certifications & Leadership

- **Certifications:** Statistical Analysis with R for Public Health (Imperial College London); Google Data Analytics Professional Certificate (Google); ALX Data Analytics (ALX).
- **Leadership:** Vice-President, HECO Club (UM6P, 2022–2024) — contributed to organizing university-wide health awareness events.